MILITARY SPECIFICATION

CONTROL-TRANSMITTER, LIQUID LEVEL, THERMISTOR TYPE, AIRCRAFT

1. SCOPE

1.1 SCOPE.- This specification covers the requirements for an aircraft thermistor type liquid level control-transmitter utilized in aircraft fuel tanks using hydrocarbon fuel.

1.2 CLASSIFICATION.- The control-transmitter shall consist of one control unit and one or more transmitter units. One single pole double throw electrical switch shall be included per transmitter unit. The control-transmitter shall be of the following types, classes, grades, powers, and ratings, as specified. (See 6.2)

Type I	Integral control and transmitter unit(s)
Type II	Separate control and transmitter unit(s)
Class 1	Contains an indicator circuit to show the control-transmitter is operable.
Class 2	Does not contain an indicator circuit
Grade A	Operating time: 1.0 second maximum from dry-to-wet 8.0 seconds maximum from wet-to-dry
Grade B	Operating time: 3.0 seconds maximum from dry-to-wet 24.0 seconds maximum from wet-to-dry
Power A	28 volts dc
Power B	115 volts, 400 cycles ac
Power C	200 volts, 400 cycles ac
Rating A	Electrical switch suitable for a 28 volt dc circuit, 10.0 amperes maximum resistive load* 10.0 amperes maximum inductive load*
Rating B	Electrical switch suitable for a 28 volt dc circuit, 5.0 amperes maximum resistive load* 2.5 amperes maximum inductive load*
Rating C	Electrical switch suitable for a 28 volt dc circuit, 1.0 ampere maximum resistive load* 0.5 ampere maximum inductive load*
Rating D	Electrical switch suitable for a 115 volt ac circuit, 2.5 amperes maximum resistive load* 2.5 amperes maximum inductive load*
Rating E	Electrical switch suitable for a 200 volt ac circuit, 1.5 amperes maximum resistive load* 1.5 amperes maximum inductive load*

*Steady state values

FSC 6680

AF Hooven-Dayton Co., 19 January 1961 200

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

P-S-661	Solvent; Dry-Cleaning
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Boxes, Fiber
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-676	Boxes, Set-Up, Paperboard
PPP-C-843	Cushioning Material, Cellulosic
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof - For Packaging and Sealing

Military

JAN-P-100	Packaging and Packing For Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods Of
MIL-B-138	Boxes, Wood, Fiberboard-Lined For Overseas Shipment (For Weight Contents Not Exceeding 500 Pounds)
MIL-S-3136	Standard Test Fluids Hydrocarbon and Isooctane
MIL-C-5015	Connectors, Electrical, "AN" Type
MIL-D-5028	Drawings and Data Lists; Preparation Of Manufacturers' (For Production Aeronautical and Associated Equipment)
MIL-W-5086	Wire, Electrical, 600 Volt, Copper, Aircraft
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-P-5315	Packing, "O" Ring, Hydrocarbon Fuel Resistant
MIL-G-5572	Gasoline, Aviation: Grades 80/87, 91/96, 100/130, 115/145
MIL-F-5624	Fuel, Aircraft Turbine and Jet Engine, Grades JP-3, JP-4, and JP-5
MIL-P-6906	Plates, Information and Identification
MIL-F-7024	Fluids, Calibrating, For Aircraft Fuel System Components

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MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-E-7894	Electric Power, Aircraft, Characteristics Of
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-T-9107	Test Reports; Preparation Of
MIL-L-10547	Liners, Case, Waterproof
MIL-N-25027	Nut, Self-Locking, 250°F, 550°F, and 800°F

STANDARDS

Federal

Federal Test Method Standard

No. 151	Metals; Test Methods

Federal Standard

No. 595	Colors

Military

MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS20995	Wire, Lock
MS29513	Packing "O" Ring Hydrocarbon Fuel Resistant
MS33586	Metals, Definition of Dissimilar
MS33588	Nuts and Plate Nuts, Self-Locking, Functional Limitations Of

Air Force-Navy Aeronautical

AN3179	Inductor - 30 Volt Laboratory Test
AND10049	Bosses - Fluid Connection, Internal Straight Thread
AND10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread
AND10056	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seat

PUBLICATION

Air Force-Navy Aeronautical Bulletin

No. 143	Specification and Standards; Use of
No. 438	Age Controls For Synthetic Rubber Parts

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(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 OTHER PUBLICATIONS.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL ADVISORY COMMITTEE FOR AERONAUTICS

Technical Note 3182 Manual of the ICOA Standard Atmosphere Calculations by the NACA - May 1954

(Copies of this publication may be obtained from the Research Information Division, NACA, 1724 F Street, N.W., Washington, D. C.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 COMPONENTS.- Each control-transmitter shall be either type I or type II, with classes, grades, powers and ratings as applicable, and shall consist of one control unit and one or more transmitter units.

3.3 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with ANA Bulletin 143 except as provided in 3.3.1 and 3.3.2.

3.3.1 COMMERCIAL PARTS.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

a. They can be replaced by standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 STANDARD PARTS.- With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

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3.4 MATERIALS.- Materials and processes used by the manufacturer of the thermistor type liquid level control-transmitter shall be of high quality, suitable for the purpose, and shall conform to applicable Government specifications. Materials conforming to contractors specifications may be used provided the specifications are released by the Services and contain provision for adequate tests. The use of contractor's specifications shall not constitute waiver of Government inspection. All materials used in the control-transmitter shall be sufficiently resistant to fluids conforming to Specifications MIL-F-5624, MIL-G-5572, and MIL-S-3136, type I and III, of aromatic content from 0 to 30 percent, to assure satisfactory operation as herein defined.

3.4.1 FUNGUS-PROOF.- Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.4.2 METALS.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service. Metals shall be inspected in accordance with Federal Test Method Standard No. 151. The use of magnesium or any alloy thereof is prohibited.

3.4.2.1 DISSIMILAR METALS.- The use of dissimilar metals shall be avoided where practicable. Dissimilar metals are defined in Standard MS33586.

3.4.3 SYNTHETIC.- For control-transmitters which include parts fabricated of synthetic material in contact with fuel, manufacturers shall control subsequent batches to provide for uniformity.

3.4.3.1 AGE CONTROLS.- Age controls for synthetic-rubber parts shall comply with the requirements of ANA Bulletin 436.

3.4.4 CASTINGS.- Castings shall be of high grade quality, clean, sound and free from blow-holes, porosity, cracks, and any other defects.

3.5 DESIGN.- The design of the control-transmitter shall be in accordance with standard aircraft practices. The weight and size of the unit shall be kept to a minimum.

3.5.1 MAINTENANCE.- The design of the control-transmitter shall permit easy assembly, disassembly, location of trouble sources, and maintenance using tools and equipment normally available to service personnel. Special tools shall not be required and maintenance shall be accomplished with a minimum of training.

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3.5.2 INSTALLATION HAZARDS.- Any portion of the control-transmitter designed to be operated in or in contact with fuel or fuel vapor shall not at any time or under any conditions exceed a temperature of 400°F. Any electrical circuitry designed to be operated in or in contact with fuel or fuel vapor shall not at any time or under any conditions, including malfunctions, be capable of initiating a fire or explosion. All transmitter units must incorporate fuses or other suitable protective devices to prevent excessive thermistor heating as a result of incorrect electrical installation or malfunction. The fuses or protective devices shall be integral with the transmitter unit. Devices which can be reset should be accessible to maintenance personnel and should not require disassembly of the transmitter unit. Devices which cannot be reset shall not be readily accessible to maintenance personnel.

3.6 CONSTRUCTION.- The control-transmitter shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service use.

3.6.1 "O" RINGS.- All "O" rings shall conform to the requirements of Specification MIL-P-5315 and the dimensional requirements of Standard MS29513.

3.6.2 SERVICEABILITY.- All synthetic-rubber parts shall be readily replaceable with a minimum replacement of attaching parts.

3.7 PERFORMANCE.- The control-transmitter and individual units thereof shall satisfy the following performance requirements:

3.7.1 DIELECTRIC STRENGTH.- The control-transmitter shall have an electrical current flow, not in excess of 2 milliamperes or a breakdown of the insulation when subjected to the dielectric strength test as described in 4.4.2.

3.7.2 CALIBRATION.- The control unit power input shall not exceed 15 watts multiplied by the number of transmitter units when subjected to the calibration test as described in 4.4.3.

3.7.3 LEAKAGE.- The control-transmitter shall have no evidence of leakage found during disassembly and inspection, when subjected to the leakage test as described in 4.4.4.

3.7.4 HOT BODY TEMPERATURE.- The control-transmitter shall have no indication of 400°F upon inspection, when subjected to the hot body temperature tests as described in 4.4.5.1 and 4.4.5.2.

3.7.5 THERMAL SHOCK.- There shall be no malfunction of the control-transmitter when subjected to the thermal shock tests as described in 4.4.6.1 and 4.4.6.2.

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3.7.6 ENVIRONMENTAL.- There shall be no evidence of malfunction, deterioration, distortion or breakdown of the control-transmitter when subjected to the vibration, fungus resistance, humidity, immersion, mechanical shock, explosion-proof, and salt spray tests as described in Specification MIL-E-5272.

3.7.7 ELECTRICAL SPARK.- The control-transmitter shall function properly when subjected to the electric spark tests as described in 4.4.13.1, 4.4.13.2, and 4.4.13.3.

3.7.8 ENDURANCE.- The control-transmitter shall function properly when subjected to the endurance tests as described in 4.4.14, 4.4.14.1, 4.4.14.2, 4.4.14.3, and 4.4.14.4.

3.7.9 DISASSEMBLY AND INSPECTION.-There shall be no evidence of deterioration or undue wear when subjected to the disassembly and inspection test as described in 4.4.15.

3.7.10 FLUID DYNAMIC LOADING.- There shall be no evidence of failure or distortion when subjected to the fluid dynamic loading test as described in 4.4.16.

3.7.11 FUEL RESISTANCE.- The control-transmitter shall function properly when subjected to the fuel resistance test as described in 4.4.17.

3.7.12 WEIGHT.- The control-transmitter shall not exceed the dry weight shown on the manufacturer's drawing when subjected to the weight test as described in 4.4.18.

3.8 INTERCHANGEABILITY.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-5028 shall govern changes in the manufacturer's part numbers.

3.9 ELECTRICAL PROPERTIES

3.9.1 DC OPERATING VOLTAGE.- Equipment intended for use in 28 volt dc electrical systems shall perform satisfactorily over a range of voltage from 20 to 29 volts. The equipment shall be functionally operable at 18 volts. Additional information on the characteristics of aircraft electrical power is provided in Specification MIL-E-7894.

3.9.2 AC OPERATING VOLTAGE AND FREQUENCY. - Equipment is intended for use in 115 volt, 400 cycle electrical systems shall operate satisfactorily at any applied frequency between 380 and 420 cycles and at any applied line to line terminal voltage between 105 and 125 volts. Equipment intended for use in 200 volt, 400 cycle electrical systems shall operate satisfactorily at any

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applied frequency between 380 and 420 cycles and at any applied line to line terminal voltage between 190 and 220 volts. Additional information on the characteristics of aircraft electrical power is provided in Specification MIL-E-7894.

3.9.3 WIRE.- Internal wire other than windings shall be in accordance with Specification MIL-C-5086.

3.9.4 ELECTRICAL CONNECTORS.- The electrical connectors shall be in accordance with Specification MIL-C-5015. The size and type of connectors shall be as specified on the manufacturer's drawing.

3.9.4.1 Grounding.- Each electrical connector used in the control-transmitter shall incorporate a ground return terminal. Circuits shall not be internally grounded.

3.9.5 INDICATOR CIRCUIT.- A single pole double throw electrical switch shall be provided to indicate that the control-transmitter is capable of detecting the absence or presence of fuel. The electrical switch rating shall be the minimum applicable rating specified in 1.2.

3.9.6 INDUCTIVE ELECTRICAL LOADING.- The inductive electrical loading of dc electrical contacts during test shall be provided through the use of an inductor in accordance with Standard AN3179. The inductive electrical loading of ac electrical contacts during test shall be provided through the use of a power factor of 50 percent.

3.10 THREADED CONNECTIONS

3.10.1 PIPE THREADS.- Pipe threads shall not be used on fuel system components, except for permanent closures in which case the pipe threads shall conform to Specification MIL-P-7105.

3.10.2 SCREW THREADS.- Machine screw threads shall conform to Specification MIL-S-7742.

3.10.3 LOCKING OF PARTS.- All threaded parts shall be positively locked by safety wiring, self locking nuts in conformance with Specification MIL-N-25027, cotter pins, or other approved methods. Safety wire shall have a minimum diameter of 0.032 inch and shall conform to Standard MS 20995. Self-locking nuts shall be used in accordance with Standard MS33588. Staking and the use of lock-washers is not permitted.

3.10.4 BOSSES AND FITTING ENDS. - Bosses and fitting ends shall conform to Standards AND10049, AND10050, or AND10056, as applicable.

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3.11 TEMPERATURE RANGE.- The control-transmitter shall operate over a fuel temperature and ambient temperature range from -67°F to +158°F.

3.12 PRESSURE RANGE.- The control-transmitter shall operate over a pressure range from 30 psi to a pressure equivalent to 80,000 feet altitude.

3.13 DIMENSION. - The control-transmitter shall conform to the manufacturer's drawings, as approved by the procuring activity.

3.14 COLOR IDENTIFICATION.- Each control-transmitter shall be marked gloss-red, conforming to Federal Standard 595, color 11105. The marking shall consist of a band 1/4 inch wide, by 1 inch long on each side or around the component, a 45-degree arc 1/4 inch wide on both sides of this component, or the entire component may be colored. The color shall be permanent, and when applied to parts that may come in contact with fuel the color shall not soften, chip, crack, or cause the fuel to be affected in any way.

3.15 FINISH

3.15.1 ANODIZING.- All aluminum -alloy parts shall be anodized in accordance with Specification MIL-A-8625, or adequately treated in some other acceptable manner for corrosion prevention.

3.15.2 PROTECTIVE TREATMENT.- Steel parts, other than corrosion-resisting steel, not in moving contact, shall be cadmium plated in accordance with Specification QQ-P-416, type II. Wearing surfaces shall be chromium plated in accordance with Specification QQ-C-320 or adequately treated in some other acceptable manner for corrosion prevention, where galling of cadmium plating would cause malfunctioning. Brass, bronze, and copper-alloy parts shall be cadmium plated, or adequately treated in some other acceptable manner for corrosion prevention. Stainless-steel parts shall be passivated.

3.15.3 PAINT.- Paint shall not be used except for color code identification.

3.6. IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accord with MIL-STD-130.

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3.16.1 NAMEPLATE.- A nameplate conforming to Specification MIL-P-6906, with the following special markings, shall be added to the individual units as applicable:

a. Control-Transmitter Unit, Type I,
Specification MIL-C-25855
Class_____Grade_____
Power_____Rating_____

b. Control Unit, Type II,
Specification MIL-C-25855
Class_____Grade_____
Power_____Rating_____

c. Transmitter Unit, Type II,
Manufacturer's Part No.----------
Manufacturer's name or trade-mark
U.S.

3.16.2 SYNTHETIC RUBBER PARTS.- Equipment and assemblies containing synthetic rubber parts used in aircraft fuel systems shall be marked in accordance with ANA Bulletin 438.

3.17 WORKMANSHIP.- All details of workmanship shall be sufficiently high grade to insure satisfactory operation and service life. Parts shall not contain sharp edges, burrs, loose chips, dirt or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS.- The inspection and testing of the control-transmitter shall be classified as follows:

a. Acceptance tests.....4.2
b. Preproduction tests.....4.5

4.2 ACCEPTANCE TESTS. - Acceptance tests shall consist of:

a. Individual test
b. Periodic sampling plans and tests

4.2.1 INDIVIDUAL TESTS.- Each control-transmitter shall be subjected to the following tests as described in 4.4 TEST METHODS:

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a. Examination of product 4.4.1
b. Dielectric strength.....4.4.2
c. Calibration.....4.4.3
d. Leakage.....4.4.4

4.2.2 PERIODIC SAMPLING PLANS AND TESTS.- One control-transmitter shall be selected from each 200 or fraction thereof produced and subjected to the tests listed below and described under 4.4. TEST METHODS:

a. Hot body temperature.....4.4.5
b. Thermal shock.....4.4.6
c. Vibration.....4.4.7

4.2.2.1 REJECTION AND RETEST.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.2.2.2 INDIVIDUAL TESTS MAY CONTINUE.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of the specification.

4.2.3 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully adivise the procuring activity of all defects likely to be found and methods of correcting them.

4.3 TEST CONDITIONS

4.3.1 ATMOSPHERIC.- Unless otherwise specified, all tests required by this specification shall be at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 60° to 90°F and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values. proper allowance shall be made for the change in instrument reading.

4.3.2 ALTITUDE.- Altitude pressures shall be determined from NACA Technical Note 3182, Standard Atmosphere Calculations.

4.3.3 ATTITUDE.- Unless otherwise specified, all tests shall be conducted with the component installed in the normal mounting attitude.

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4.3.4 CLEANING.- Prior to testing the control-transmitter all parts normally in contact with the fuel or fuel vapors shall be thoroughly cleaned to remove all lubricants, protective coatings, and foreign matter.

4.3.5 TEST FLUID.- Unless otherwise specified herein, the fluid used for testing shall be conforming to Specifications MIL-G-5572, MIL-F-5624, MIL-F-7024, or test fluid conforming to Specification MIL-S-3136, type I and III. Other fluids such as fluids conforming to Specification P-S-661 may be used only with the specific approval of the procuring activity.

4.3.6 TIME AND ADJUSTMENT.- The control-transmitter shall detect the presence and absence of fuel within the time requirements as specified in 1.2, which results in the opening and closing of electrical circuits. The control-transmitter shall not require any adjustments or replacement of parts during the course of testing.

4.4 TEST METHODS

4.4.1 EXAMINATION OF PRODUCT.- Each control-transmitter submitted for acceptance shall be carefully examined for material, size, workmanship, and dimensional requirements to determine conformance with the requirements of this specification.

4.4.2 DIELECTRIC STRENGTH.- The applicable test voltage as specified in table I shall be applied between circuits and between each circuit and the metal frame of the control-transmitter. Electrical current flow in excess of 2 milliamperes or breakdown of insulation shall constitute failure.

TABLE I
Test Voltage

Normal Voltage	Test Voltage (Peak)	Time
28 V DC	1000V	1 min 1/
115/200V 400 cyc. AC	1500V	1 min 1/

1/ For individual tests this time may be reduced to 1 sec.

4.4.3 CALIBRATION.- With minimum normal line voltage applied and after a 2-minute stabilization period, the control-transmitter shall not exceed the applicable operating times specified in 1.2. For control-transmitters intended for use in 28 volt dc systems, the switch shall be functionally operable at 18 volts dc. Control-

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transmitters intended for use in ac system shall be functionally operable at the minimum normal line voltage specified in Specification MIL-E-7894. All calibration tests shall be performed using fuel conforming to Specification MIL-F-5624, grade JP-4. The maximum rated resistive load shall be imposed on all electrical contacts. The power input to each control unit shall be measured using maximum normal line voltage. The control unit power input shall not exceed 15 watts multiplied by the number of transmitter units.

4.4.4 LEAKAGE.- The leakage test is applicable to those sealed portions of the control-transmitter designed to operate within the fuel or fuel vepors as well as any mounting or sealing arrangements designed to act as a barrier to fuel or fuel vapors. A test fluid pressure from 0 to 30 psi in 5 psi increments shall be applied to the sealed portion or barrier being tested. Any leakage occurring within a 5-minute waiting period, commencing with each application of pressure, shall constitute failure. Air suction pressure from 0 to 6 psi in 2 psi increments shall be applied to the sealed portion or barrier being tested. Any leakage occurring within a 5-minute waiting period, commencing with each application of pressure, shall constitute failure. For those portions of the control-transmitter which are inaccessible for observation, any evidence of leakage found during disassembly and inspection shall constitute failure.

4.4.5 HOT BODY TEMPERATURE

4.4.5.1 EXTREME OPERATION.- The hot body or bodies of the control-transmitter designed to be operated in or in contact with fuel or fuel vapors shall be coated with a thin coating of a suitable temperature sensitive paint calibrated for detecting a temperature of 400°F. The control-transmitter shall be in a completely assembled condition prior to testing. With the hot body or bodies heated and stabilized in circulating ambient air having a temperature of 160° ± 2°F, the maximum supply voltage shall be applied to the control-transmitter for not less than 2 minutes. Upon inspection of the temperature sensitive point, any indication of 400°F operation shall constitute failure. Other methods of measuring hot body temperature may be used at the discretion of the activity responsible for approval.

4.4.5.2 INCORRECT INSTALLATION OR MALFUNCTION.- Each combination of a thermistor bead and its protective device shall be subjected to this test. The thermistor bead and its protective device shall be connected in series. Both components shall be coated with a thin coating of a suitable temperature sensitive paint calibrated

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for detecting a temperature of 400°F. The components shall be heated and stabilized for 2 minutes in circulating ambient air having a temperature of 160° ± 2°F. Maximum line voltage shall be applied instantaneously. Upon inspection of the temperature sensitive paint, any indication of 400°F operation shall constitute failure.

4.4.6 THERMAL SHOCK

4.4.6.1 COLD AIR-HOT FUEL SHOCK.- The control-transmitter shall be exposed to ambient air at -65° ± 4°F for 1 hour with maximum line voltage applied. At the end of the soak period and while still at -65°F, the calibration test shall be performed using a suitable test fuel which has been preheated to 135° ± 4°F. Three complete cold air-hot fuel cycles are required during which there shall be no failure or malfunction.

4.4.6.2 HOT AIR-COLD FUEL SHOCK.- The control-transmitter shall be exposed to ambient air at +160° ± 4°F for 1 hour with maximum line voltage applied. At the end of the soak period and while still at +160°F, the calibration test shall be performed using a suitable test fuel which has been cooled to -65° ± 4°F. Three complete hot air-cold fuel cycles are required during which there shall be no failure or malfunction.

4.4.7 VIBRATION.- The vibration test shall be conducted in accordance with Specification MIL-E-5272, procedure I. Functional operation shall be checked and found to be satisfactory during each phase of testing. Any erratic or inadvertent operation shall be detected by means of an oscilloscope. At the end of this test there shall be no evidence of failure or distortion and the calibration test shall be performed.

4.4.8 FUNGUS RESISTANCE.- The control unit shall be subjected to this test in accordance with the requirements of Specification MIL-E-5272, procedure I. After completion of the test, the complete control-transmitter shall be subjected to the calibration test.

4.4.9 HUMIDITY.- The control-transmitter shall be subjected to the humidity test in accordance with Specification MIL-E-5272, procedure I. There shall be no evidence of malfunction or deterioration that might adversely affect performance. At the end of the test period, the calibration test shall be performed.

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4.4.10 IMMERSION.- The control-transmitter shall be subjected to the immersion test in accordance with Specification MIL-E-5272, procedure I. Leakage as defined in the test procedure shall constitute failure. At the end of the test period, the calibration test shall be performed.

4.4.11 MECHANICAL SHOCK.- The control-transmitter shall be subjected to an equipment operation shock test in accordance with Specification MIL-E-5272, procedure II. Maximum line voltage shall be applied during test. At the end of this test there shall be no evidence of failure or distortion and the calibration test shall be performed.

4.4.12 SALT SPRAY.- The control-transmitter shall be subjected to a salt spray test in accordance with Specification MIL-E-5272. At the end of the test period, the salt deposits shall be removed with warm water, dried, and the calibration test shall be performed. Any evidence of corrosion which might adversely affect the control-transmitter shall be cause for rejection.

4.4.13 ELECTRICAL SPARK

4.4.13.1 DETERMINING MALFUNCTIONS.- A combination of as many as three likely simultaneous failures or malfunctions shall be determined which would produce the greatest spark energy at those elements intended for usage in contact with fuel or fuel vapors. One of the failures or malfunctions shall consist of the open circuit at which the sparking would occur. Failures or malfunctions shall be considered to be open circuits, short circuits, and partially opened or shorted circuits. An open fuse or protective device shall not be considered as a malfunction during this determination.

4.4.13.2 MODIFICATION.- After the determination of 4.4.13.1 the control-transmitter will be modified to incorporate the failures or malfunctions. If an inductor or inductors are employed in the circuit, the electrical energy storage capability of the circuit will be doubled by the addition, in parallel, of an inductor with identical characteristics to the inductor being used, that is, same inductance, resistance, and eddy current loss. It is permissible to compensate for the decrease in resistance resulting from the paralleling of inductors.

4.4.13.3 SPARK TEST.- After modifications per 4.4.13.2, an explosion proof test will then be conducted in accordance with Specification MIL-E-5272, procedure I, with the following exceptions:

The control-transmitter shall be mounted outside of the explosion chamber with lead wires extending from the control-transmitter to a rotating spark gap device located within the

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explosion chamber. The lead wires will carry the sparking potential created by the simulated malfunctions. The rotating spark gap device shall consist of a No. 29 (AWG) untinned copper wire fastened radially to a shaft rotating at 20 to 30 rpm. This length of wire shall impinge on another No. 29 (AWG) wire at right angles to it and in such a manner that a sudden "snap" break action is accomplished when the contact between the wires is broken. With maximum voltage applied to the system, with the rotating spark gap device in operation, and with the use of the explosion test procedure as specified in Specification MIL-E-5272, any explosion shall constitute a failure.

4.4.14 ENDURANCE.- The control-transmitter shall be operated for a minimum of 100,000 complete cycles. The maximum rated resistive load shall be applied for 90,000 cycles and the maximum rated inductive load shall be applied for 10,000 cycles. The endurance test shall consist of the following:

a. Room temperature (75 percent of total)
b. Altitude (15 percent of total)
c. Low temperature (5 percent of total)
d. High temperature (5 percent of total)

The cycling may be accomplished automatically by utilizing the control-transmitter itself to start and stop each cycle. In this case, the cycling rate would be determined by the natural response time of the system at rated voltage. A cycle is defined as one dry-to-wet operation and one wet-to-dry operation.

4.4.14.1 ROOM TEMPERATURE ENDURANCE.- The room temperature endurance test shall be conducted at room temperature using a suitable test fluid. At the end of the test period, the calibration test shall be performed.

4.4.14.2 ALTITUDE ENDURANCE.- The altitude endurance test shall be conducted at a simulated altitude of 80,000 feet, except that the ambient temperature shall be room temperature. Fuel conforming to Specification MIL-F-5624, grade JP-4, shall be used and shall not be weathered. The calibration test shall be performed at the beginning and end of the test period.

4.4.14.3 LOW TEMPERATURE ENDURANCE.- The low temperature endurance test shall be conducted using a fluid and ambient temperature of -67° ± 2°F. A suitable test fluid shall be used. At the end of the test period, the calibration test shall be performed.

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4.4.14.4 HIGH TEMPERATURE ENDURANCE.- The high temperature endurance test shall be conducted using a fluid and ambient temperature of 158° ± 2°F. A suitable test fluid shall be used. At the end of the test period, the calibration test shall be performed.

4.4.15 DISASSEMBLY AND INSPECTION.- The control-transmitter shall be disassembled and inspected. There shall be no evidence of deterioration or undue wear.

4.4.16 FLUID DYNAMIC LOADING.- That portion of the control-transmitter projecting into the fuel and subjected to fluid force as a result of fuel movement shall be tested as follows: A mechanically simulated fluid loading of 5 pounds per square inch of projected area shall be alternately applied on opposite sides of the projecting member normal to the major axis at a rate of 12 cycles per minute for a 10-hour period. At the end of the test period, there shall be no evidence of failure or distortion and the calibration test shall be performed.

4.4.17 FUEL RESISTANCE.- The control-transmitter shall be subjected to a fuel resistance and extreme temperature test in accordance with table II. In certain cases, the fuel resistance portion of the test may be waived subject to the approval of the procuring activity.

4.4.18 WEIGHT.- The control-transmitter shall be weighed in a dry condition. The weight shall not exceed the weight shown on the manufacturer's drawing.

4.5 PREPRODUCTION TESTING

4.5.1 PREPRODUCTION TEST SAMPLES TESTED BY THE CONTRACTOR.- The contractor shall subject the control-transmitter to the preproduction test specified in 4.5.5.

4.5.2 PREPRODUCTION TEST REPORT.- After the contractor completes the preproduction tests, he shall prepare a preproduction test report according to Specification MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.5.3 PREPRODUCTION TEST SAMPLE FOR THE PROCURING ACTIVITY.- Three untested control-transmitters shall be submitted for each manufacturer's part number upon which approval is requested.

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TABLE II

Fuel Resistance and Extreme Temperature Test Schedule

Test	Fuel Resistance					Extreme Temperatures	
Period 1/	Initial Drying	Phase I soak 2/	Phase I dry	Phase II soak 2/	Phase II dry	Low Temperature	High Temperature
Test Fluid	None	MIL-S-3136 Type III	None	MIL-S-3136 Type I	None	MIL-S-3136 Type I	MIL-S-3136 Type III
Period duration	168 hours (7 days)	168 hours (7 days)	4 hours	504 hours (21 days)	4 hours	72 hours (3 days)	72 hours (3 days)
Ambient and test fluid temperatures	158° ± 2°F	Room	158° ± 2°F	Room	158° ± 2°F	-67°F ± 2°F	160° ± 5°F
Operation or tests during period	None	Actuate component at least twice per day	None	Actuate component at least twice per day	None	None	None
Operation or test immediately after period	Conduct leakage and calibration tests 3/	Conduct leakage test using MIL-S-3136 type III fluid	Conduct leakage and calibration tests	Conduct leakage test using MIL-S-3136 type I fluid	Conduct leakage and calibration tests	(a) Conduct calibration test at test temperatures. (b) Direct an air flow of 5 ft/sec at the transmitter unit while at the test temperatures. There shall be no erratic or inadvertent operation. (c) Any additional calibration test shall be run after coating the transmitter unit with a frost thickness of approximately 1/32 of an inch.	Conduct calibration test at test temperatures

1/ Each period shall follow immediately after the preceding one in the order noted.

2/ During periods of soaking in the test fluid, the component shall be maintained in such a manner as to insure complete contact of all parts intended for use in fuel or fuel vapors.

3/ Leakage shall be kept within the allowable limits of this specification except for the first 15 minutes of the leakage tests conducted after the three drying periods.

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4.5.4 PREPRODUCTION DRAWINGS.- One complete set of detail and assembly drawings shall be submitted including any electrical wiring or circuit diagrams. The assembly drawings shall contain a sufficient number of cross section views to facilitate a complete understanding of assembly and functional operation. Part numbers shall be shown on assembly drawings. The material and finish information for each part shall be shown on the respective part drawing and shall conform to the requirements of this specification. The manufacturer's material or finish designations will not be accepted unless accompanied by complete descriptive information.

4.5.5 PREPRODUCTION TESTS.- The preproduction tests of the control-transmitters shall consist of the following tests in the order listed, as described under "TEST METHODS".

a. Tests for Sample No. 1

(1) Examination of product.....4.4.1
(2) Hot body temperature 1/.....4.4.5
(3) Dielectric strength.....4.4.2
(4) Fungus resistance.....4.4.8
(5) Electrical spark.....4.4.13
(6) Disassembly and inspection.....4.4.15
(7) Weight.....4.4.18

b. Test for Sample No. 2

(1) Examination of product.....4.4.1
(2) Dielectric strength.....4.4.2
(3) Immersion.....4.4.10
(4) Calibration.....4.4.3
(5) Mechanical shock.....4.4.11
(6) Endurance.....4.4.14
(7) Fluid dynamic loading 2/.....4.4.16
(8) Salt spray.....4.4.12
(9) Disassembly and inspection.....4.4.15
(10) Weight.....4.4.18

c. Test for Sample No. 3

(1) Examination of product.....4.4.1
(2) Dielectric strength.....4.4.2
(3) Leakage.....4.4.4
(4) Calibration.....4.4.3
(5) Fuel resistance.....4.4.17
(6) Thermal shock.....4.4.6
(7) Vibration.....4.4.7
(8) Humidity.....4.4.9
(9) Disassembly and inspection.....4.4.15
(10) Weight.....4.4.18

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1/ If the method of conducting the hot body temperature test will render the control-transmitter unsatisfactory for the remaining tests, this test shall be conducted on a separate additional test sample.

2/ To be conducted at the discretion of the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 LEVELS OF PRESERVATION AND PACKAGING

5.1.1 LEVEL A.- Control-transmitters shall be preserved in accordance with Specification MIL-P-116, method IA and packed in containers meeting Specification PPP-B-566, PPP-B-676. For domestic shipment containers meeting Specification PPP-B-636 may be used. Unit quantities shall be one each.

5.1.2 LEVEL C.- Control-transmitters shall be preserved and packaged in accordance with manufacturer's commercial practice.

5.2 LEVELS OF PACKING

5.2.1 LEVEL A.- Control-transmitters preserved and packaged to meet 5.1.1 shall be packed in exterior type shipping containers meeting Specification PPP-B-591, PPP-B-621, PPP-B-636, MIL-B-138, or PPP-B-601. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to Specification MIL-L-10547 and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the unit, intermediate, or exterior container conforms to Specification PPP-B-636 and is sealed at all joints and seams, including manufacturer's joint, with tape conforming to Specification PPP-T-60, type I, class 1.

5.2.2 LEVEL C.- Packages which require over-packing for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or Regulations of other common carriers as applicable to the mode of transportation.

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5.3 PHYSICAL PROTECTION.- Cushioning, blocking, bracing, and bolting as required shall be in accordance with Specification JAN-P-100 except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of Specification JAN-P-100 shall be waived when preservation, packaging, and packing of the item is for immediate use or when drop tests of Specification MIL-P-116 are applicable. Cushioning material shall be in accordance with, or equal to, PPP-C-843.

5.4 MARKING.- Interior and exterior containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows:

a. For type I:

Control-Transmitter, Liquid Level, Thermistor Type, Aircraft
Specification MIL-C-25855
Type I, Class_____, Grade_____, Power_____, Rating_____.
Federal Stock No._____

b. For type II:

(1) Control Unit
Control, Liquid Level, Thermistor Type, Aircraft
Specification MIL-C-25855
Type II, Class_____, Grade_____, Power_____, Rating_____.
Federal Stock No._____

(2) Transmitter Unit
Transmitter, Liquid Level, Thermistor Type, Aircraft
Specification MIL-C-25855, Type II,
Federal Stock No._____

6. NOTES

6.1 INTENDED USE.- The aircraft thermistor type liquid level control-transmitter is used for the purpose of controlling the operation of pumps and valves, or actuating warning lights, buzzers, alarms, or other signalling devices. It is intended to mount in aircraft fuel tanks using hydrocarbon fuel with an ambient temperature range from -67°F to +158°F and a pressure range from 30 psi to a pressure equivalent to 80,000 feet altitude.

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6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Type, class, grade, power, and rating required.

b. Selection of applicable levels of preservation and packaging, and packing.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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